MIL-G-36805
12 November 1971

MILITARY SPECIFICATION

GRADUATE, LIQUID, LABORATORY, Polypropylene

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers polypropylene graduates of a grade suitable for pharmaceutical use.

1.2 Classification. Graduates shall be of the following sizes, as specified, for the appropriate stock number and item identification:

Size 1 - Nominal capacity 125 milliliters (ml.) and 4 ounces (oz.).
Size 2 - Nominal capacity 250 ml. and 8 oz.
Size 3 - Nominal capacity 500 ml. and 16 oz.
Size 4 - Nominal capacity 1000 ml. and 32 oz.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6640

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

Lange's Handbook of Chemistry

(Application for copies should be addressed to McGraw-Hill Book Company, 330 West 42nd Street, New York, New York 10036.)

3. REQUIREMENTS

3.1 Material. Material shall be translucent polypropylene.

3.2 Construction. Graduates shall be conical and rigid, and shall be provided with a pouring lip and base. The walls shall be a minimum of 3/64 inch thick. The inside measurement from the bottom of the graduates to the capacity graduation shall be not less than twice the inside diameter at the capacity graduation. Interior of graduate shall be smooth.

3.2.1 Base. The base shall be octagonal and perpendicular to the axis of the graduate, and shall be of such dimension that the empty graduate shall not fall when tested as specified in 4.4.1.

3.2.2 Calibration. Graduates shall be calibrated to deliver the indicated volume at 20° Centigrade (C.) (68° Fahrenheit [F.]).

3.2.3 Scales. Graduates shall have two adjacent scales separated by a line; one scale shall be in milliliters and the second scale in ounces.

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3.2.4 Graduated intervals. Graduates may have an initial interval that is not graduated, equal to not more than one-quarter of the capacity of the graduates for ounces, and one-fifth of the capacity for milliliters. Except for this initial interval, the values of all graduated intervals shall be the same. Nominal capacities, graduation ranges, and main and intermediate graduations shall be as shown in table I.

Table I. Design Details

Size	Nominal capacity	To be graduated between	Intermediate graduation*	Main graduation*
	(ounces)	(ounces)	(ounces)	(ounces)
1	4	1 and 4	1/2	1
2	8	2 and 8	1/2	1
3	16	4 and 16	1	2
4	32	8 and 32	1	4
	(milliliters)	(milliliters)	(milliliters)	(milliliters)
1	125	225 and 125	5	25
2	250	50 and 250	10	50
3	500	100 and 500	25	100
4	1000	200 and 1000	25	100

* Graduated intervals in the initial one-quarter or one-fifth may be smaller than those shown in table I. All graduated intervals shall meet the requirements specified in 3.3.

3.2.5 Graduation markings. Graduation markings shall be raised and legible. Each of the graduation lines shall be perpendicular to the vertical axis of the graduate and scale separation line. Main graduation lines shall be numbered with Arabic numerals. Intermediate graduation shall not be numbered, except for size 3, the 3-, 5-, and 7- oz., and 150- and 250- ml. graduations may be numbered. The main graduations shall be longer than the intermediate graduations. Main graduation numbers shall be uniformly positioned at the outer end of the applicable graduation line.

3.3 Tolerances. Tolerances shall be as shown in table II (the tolerance to be applied at any graduation shall be determined by the inside diameter of the graduate at the graduation in question).

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Table II. Tolerances for Graduates*

Inside diameter of graduate (inches)	Tolerance (plus or minus) (drams)	(minims)	Inside diameter of graduate (millimeters)	Tolerance (plus or minus) (milliliters)
13/16 to 1-1/16		6	21 to 26	0.4
1-1/16 to 1-5/16		10	26 to 31	0.6
1-5/16 to 1-9/16		15	31 to 36	0.8
1-9/16 to 1-13/16		20	36 to 41	1.1
1-13/16 to 2-1/16		30	41 to 46	1.4
2-1/16 to 2-5/16		40	46 to 51	1.8
2-5/16 to 2-9/16		50	51 to 56	2.2
2-9/16 to 2-13/16	1	5	56 to 61	2.8
2-13/16 to 3-1/16	1	20	61 to 66	3.4
3-1/16 to 3-5/16	1	35	66 to 71	4.1
3-5/16 to 3-9/16	1	50	71 to 76	4.8
			76 to 81	5.6
			81 to 86	6.4

* Larger tolerance shall apply to duplicating diameters.

3.4 Capacity. The graduate shall meet the requirements specified in 3.3, when tested as specified in 4.4.2.

3.5 Heat resistance. The graduate shall not soften, deform, crack, or become opaque, when tested as specified in 4.4.3.

3.6 Resistance to dripping. The outside of the graduate shall show no evidence of water when the graduate is tested as specified in 4.4.4.

3.7 Identification marking. The name or registered trademark of the manufacturer, the calibration means (to deliver), and temperature (20° Centigrade [C.]) shall be marked on the graduates with raised marking similar to the graduation marking. The calibration means and temperature may be abbreviated.

3.8 Workmanship. Workmanship shall be first class throughout. Graduates shall be free of defects listed in table V and other defects which detract from their appearance or may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, and auditory, investigation *without the use of special laboratory appliances or procedures*) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table III. Unit of product shall be one graduate.

Table III. Sampling for examination

	Inspection level	AQL (percent defective
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-1	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table IV. Unit of product shall be the graduate; acceptance number shall be zero, and inspection level shall be S-1.

Table IV. Sampling for tests

Characteristic	Requirement	Test Procedure
Stability	3.2.1	4.4.1
Capacity	3.4	4.4.2
Heat resistance	3.5	4.4.3
Resistance to dripping	3.6	4.4.4

4.3 Examination. The graduates shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the defects listed in table V, but shall not be limited to those defects.

Table V. Classification of defects

Categories	Defects
Major	
101	Graduation and calibration markings not correct or legible.
102	Graduate not free of deformed areas.
103	Graduate not free of opaque areas.
104	Graduate not free of tears.
105	Graduate not free of cracks.
106	Graduate not free of sharp edges.
107	Interior of graduate not smooth.
Minor	
201	Graduate not free of crazing.
202	Edges of graduate not free of feathering.
203	Exterior of graduate not smooth (with the exception of markings and mold).
204	Manufacturer's identification not complete, correct or legible.

4.3.2 Dimensional examination. The graduates shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

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4.4.1 Stability. The graduate shall be placed on a smooth surface which makes a 15° angle with the horizontal.

4.4.2 Capacity. Graduates shall be thoroughly cleaned, rinsed with distilled water and dried. A minimum of three graduations of the milliliter range and a minimum of three graduations of the ounce range shall be tested with distilled water. Total capacity (1000 ml. and 32 ounce graduations of 1000 ml. graduate) and 2/3 and 1/3 capacities shall be tested. Where 2/3 and 1/3 capacities are not marked, nearest graduation mark shall be tested. Graduates and distilled water shall be allowed to stand at room temperature for at least 1/2 hour. Room temperature shall be 20° C. ± 5°. Water shall be added to the graduate until the applicable graduation has been reached. The water shall be delivered into a tared container. (If water is left in the graduate, it is evidence that the graduate is not clean; therefore, result shall be disregarded, the graduate recleaned, rinsed and dried.) The container and water shall be weighed, the weight recorded, the temperature of the water taken and recorded. Weight of the water shall be determined by subtracting the weight of the container from the weight of the water and container. Volume delivered shall be determined by dividing the weight of the water by the relative density of the water at the testing temperature. Relative density shall be as indicated in Table for Relative Density and Volume of Water and of Mercury at Various Temperatures in Lange's Handbook of Chemistry. For calculations of capacities for ounce graduation, ounces shall be converted to milliliters and 1 fluid ounce shall be considered equal to 29.5729 ml.

4.4.3 Heat resistance. The graduate shall be placed in an autoclave for at least 20 minutes at 15 pounds per square inch and 121° C. (250° F.).

4.4.4 Resistance to dripping. Graduate shall be filled with water to the nominal capacity and the water poured out through the pouring lip. (Care should be taken to prevent water from spilling on the exterior of the graduate during the filling process.)

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One graduate, of the size specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table VI shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table VI. Packaging quantities

Size	Unit package	Intermediate package	Exterior container
1	1 unit	12 units	144 units
2	1 unit	12 units	144 units
3	1 unit	12 units	144 units
4	1 unit	6 units	72 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be wrapped in tissue, and packaged in a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676 except that commercial colors are acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

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5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure of wood boxes shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Federal stock number (FSN).
c. Appropriate size and capacity.
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

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